# Standardizing nanoparticle quantification in magnetic particle imaging through an adaptive open source analysis framework

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## 1. WHAT IS MAGNETIC PARTICLE IMAGING (MPI)?

Magnetic Particle Imaging (MPI) is an emerging tomographic technique free of radiation that directly maps superparamagnetic iron oxide nanoparticles with exceptional sensitivity and zero anatomical background signal for diverse biomedical applications [1, 2, 3]. Broader translation is hindered by inconsistent data-analysis protocols, which drive user- and scanner-dependent variation in reconstruction and quantification across laboratories [4, 5].

Although MPI is volumetric, 2D projections or slices remain essential for rapid evaluation of reconstruction methods and preclinical quantification [1, 6].

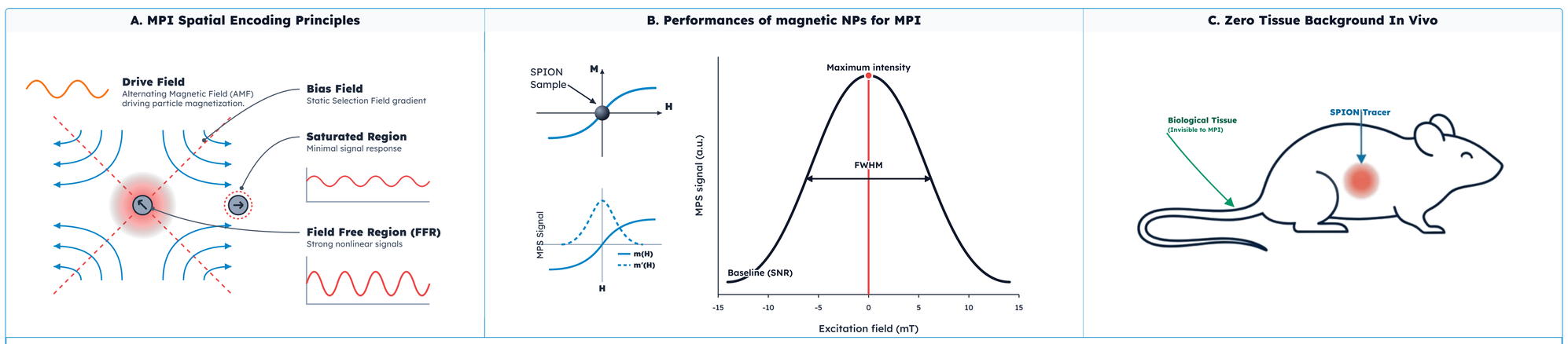
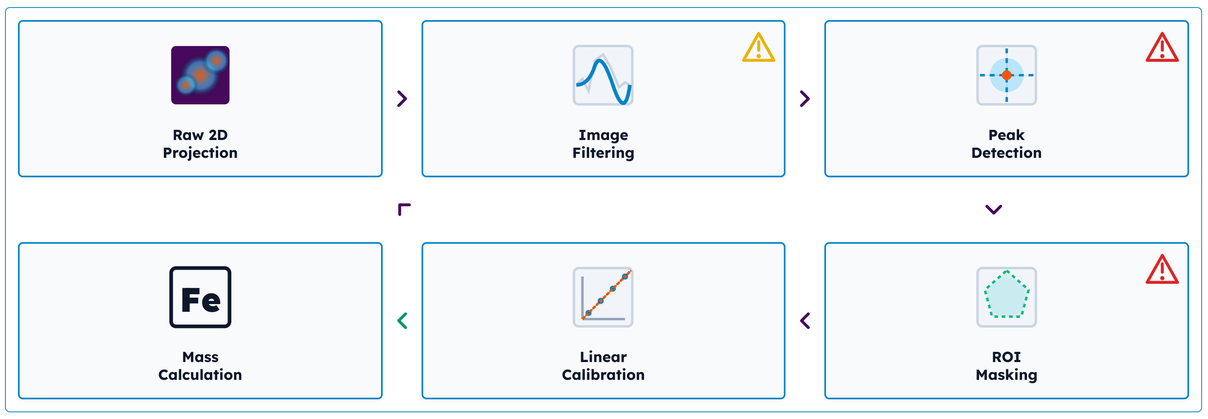



*Fig. 1: (A) Static selection-field gradient and alternating drive field: signal preferentially in the field-free region (FFR), surrounding voxels saturated. (B) Top-left: SPION magnetization (M) vs. magnetic field (H). Bottom-left: Magnetization m(H) and its derivative m'(H) (MPS signal). Right: MPS signal peak vs. excitation field, annotated with quality metrics: maximum intensity, Full Width at Half Maximum (FWHM) of the signal distribution, and baseline (SNR). (C) In vivo, tissue contributes negligible MPI background; tracer localizes to hotspots; anatomical context requires MRI or CT co-registration.*

## 2. ANALYTICAL METHODOLOGY: THE 2D CALIBRATION PIPELINE

While inherently volumetric, 2D images remain crucial for rapid iron tracking and quantification. The standard route for 2D analysis involves a sequential multi-steps data extraction workflow.




## 3. THE FUNDAMENTAL QUANTIFICATION BOTTLENECK

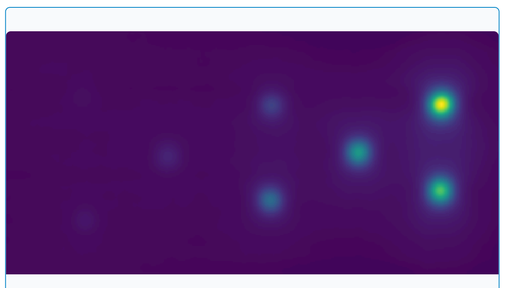


*Fig. 2: 2D MPI (Magnetic Insight Inc.) projection of eight iron oxide nanoparticle phantoms at varying concentrations. The resulting high dynamic range demonstrates the inherent unreliability of manual delineation, as human operators struggle to consistently threshold both saturated and low-intensity signals.*

Despite MPI's high sensitivity and technical potential, cross- and intra-lab validation are hindered by non-standardized workflows that are inherently time-consuming, requiring multiple scans, extensive post-processing, and highly experienced users.

- **Subjective identification:** Visual spot detection can miss or mischaracterize low-intensity samples.
- **Ambiguous delineation:** User-defined Regions of Interest (ROIs) lack consensus; small boundary shifts can substantially change extracted intensities.
- **Algorithmic rigidity:** Fixed pipelines impose a one-size-fits-all workflow that does not adapt to diverse nanoparticle properties or complex image configurations (e.g., overlapping signals, irregular spatial arrangements).
- **Compounded error:** As these factors accumulate, obtaining consistent mass estimates from identical raw data becomes difficult, and repeated human decisions at each step reduce reproducibility.

## 4. ADAPTIVE LOGIC & SYNERGISTIC BENCHMARKING

To completely eliminate laboratory and operator bias, we first systematically mapped the 2D calibration space under *in vitro* conditions. We acquired physical 2D scans of diverse SPION formulations across varying concentrations and complex spatial arrangements, including both single and multiple sample images. Crucially, every single algorithmic output was strictly validated against absolute iron mass determined via ICP-OES (our gold standard).

By evaluating combinations of over 40 classical methods and 10 machine learning models, competing across filtering, peak detection, and ROI segmentation, we benchmarked over 1.1 million unique pipeline configurations to identify the exact mathematical synergies capable of achieving >99% linearity between the ICP-OES ground truth and the iron mass quantified by the pipeline.

This linearity is measured by the coefficient of determination ($R^2$), which defines exactly how much of the observed physical data is successfully explained by the model.

This massive foundational dataset directly powers our automated framework. The system uses a "Smart Selector" to analyze raw image parameters (e.g., global SNR, peaks FWHM) and dynamically construct the optimal mathematical pipeline for that specific scan:

1. **Filtering:** Performs precise noise reduction and peak enhancement depending on specific image conditions.
2. **Peak Detection:** Selects the optimal signal regions while ignoring false positives.
3. **ROI Masking:** Adapts boundary delineation to maximize true signal area while avoiding overlap and background noise.

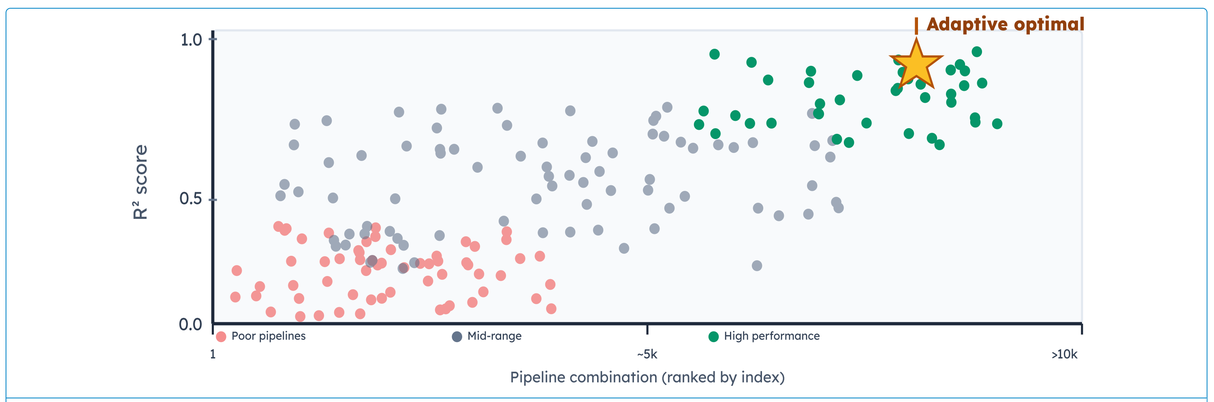



*Fig. 3: Performance landscape of >10,000 evaluated pipeline configurations against the known ground truth. This widespread distribution illustrates the massive computational effort required to identify the single 'Adaptive optimal' winner capable of achieving >99% linearity for every dataset.*

## 5. PROPOSED ADAPTIVE PIPELINE

By leveraging the optimization strategy outlined in Section 4, the framework eliminates user subjectivity and sensitivity to specific lab setups, thus enabling repeatability and cross-laboratory standardization for generic MPI 2D images.

**1. Adaptive Analysis**
Automatically selects the optimal pipeline for each dataset.

**2. Accurate Quantification**
Reduces variability and improves reliability in iron mass estimation.

**3. Open & Evolving**
Free, open-source framework, under continuous development.

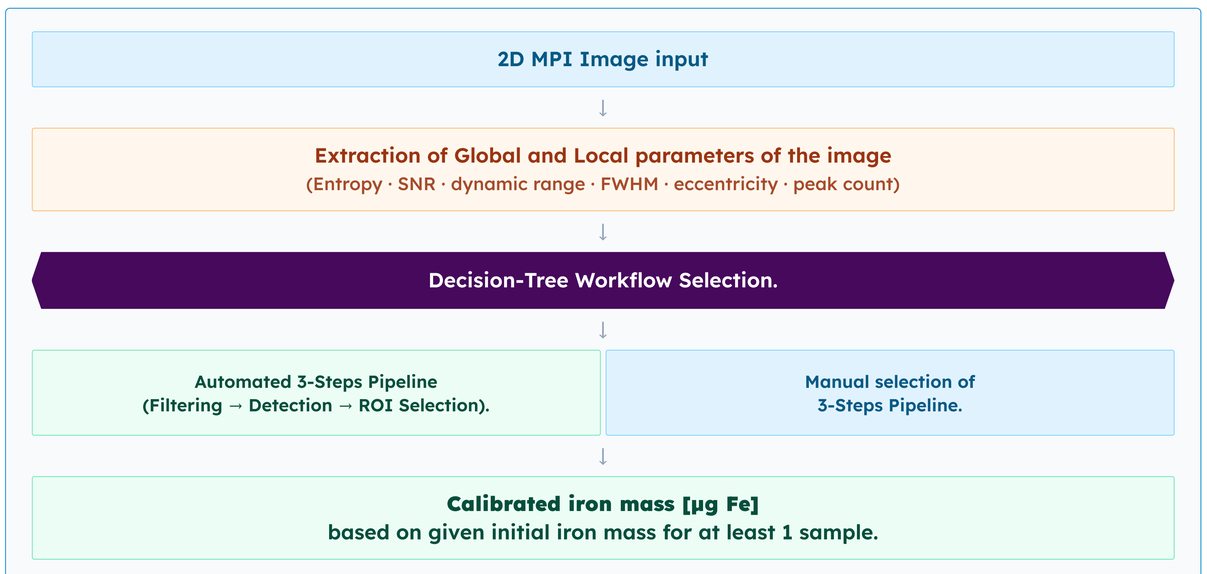



*Fig. 4: Adaptive workflow from 2D MPI input to µg Fe output: extraction of global and local image parameters drives decision-tree workflow selection, then either automated three-step pipeline (filtering → detection → ROI selection) or manual selection of the three-step pipeline for strict cross-laboratory reproducibility.*

## 6. RESULTS AND CONCLUSIONS

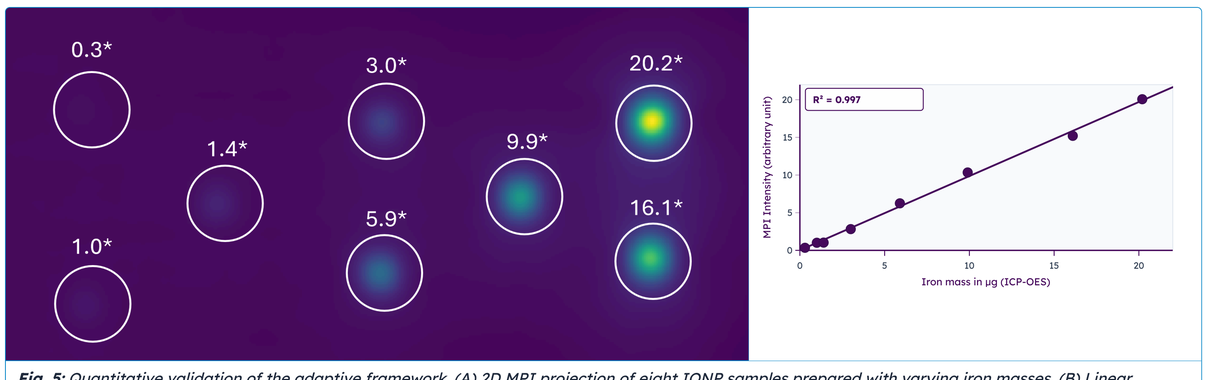



*Fig. 5: Quantitative validation of the adaptive framework. (A) 2D MPI projection of eight IONP samples prepared with varying iron masses. (B) Linear correlation between framework-quantified intensity and absolute ICP-OES reference standards. The adaptive selection yields a least-squares fit of $R^2$ = 0.997, demonstrating near-perfect agreement across the entire dynamic range.*

- **First standardization framework for absolute 2D MPI iron quantification, which enables cross-laboratory and longitudinal reproducibility.**
- **Linear ($R^2$ > 0.99) calibration curve relative to ICP-OES standards.**
- **Algorithmic selection of the calibration pipelines (validated over >1.1M instances across diverse SPION and setup conditions) that outperforms any fixed-pipeline approaches.**
- **Custom peak detection surpasses standard mathematical and ML benchmarks.**
- **Open-source code, promoting community-based build-up and further development.**

## 7. LIMITATIONS AND PERSPECTIVE

### LIMITATIONS

This computational approach is validated exclusively on *in vitro* 2D datasets. Direct translation to complex biological environments requires further investigation.

### PERSPECTIVE

- **Translational Validation:** Extending to *ex vivo* and *in vivo* scenarios characterizing highly irregular spatial iron distributions.
- **Signal Deconvolution:** Applying mathematical deconvolution models to separate overlapping Point Spread Functions (PSFs) characteristic of dense *in vivo* accumulation.
- **3D Expansion:** Adapting the algorithms natively for volumetric datasets.

## 8. REFERENCES

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**Scan for Full Version**
lorencig.github.io/#/projects/MPI